Vector-Valued Functions

ANSWER KEY

Position, velocity, and acceleration vectors

1 A particle's position is given by $\vec{r}(t) = \langle t^2, t^3 \rangle$. Find the velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$.
$\vec{v}(t) = \langle 2t, 3t^2 \rangle$. $\vec{a}(t) = \langle 2, 6t \rangle$.

2 For the same particle, find the speed at $t = 2$.
$\vec{v}(2) = \langle 4, 12 \rangle$. Speed $= |\vec{v}(2)| = \sqrt{4^2 + 12^2} = \sqrt{16 + 144} = \sqrt{160} = 4\sqrt{10}$.

Integrating vector-valued functions

3 Given $\vec{v}(t) = \langle 3t^2, 2t \rangle$ and initial position $\vec{r}(0) = \langle 1, 4 \rangle$, find $\vec{r}(t)$.
$\vec{r}(t) = \langle t^3 + C_1, t^2 + C_2 \rangle$. Using $\vec{r}(0) = \langle 1, 4 \rangle$: $C_1 = 1$, $C_2 = 4$. $\vec{r}(t) = \langle t^3 + 1, t^2 + 4 \rangle$.

Total distance traveled

4 A particle has velocity $\vec{v}(t) = \langle \cos t, \sin t \rangle$ for $t \in [0, \pi]$. Find the total distance traveled.
Speed $= |\vec{v}(t)| = \sqrt{\cos^2 t + \sin^2 t} = 1$ (constant). Distance $= \int_0^{\pi} 1\, dt = \pi$.

Vectors and parametric motion combined

5 A particle moves along $x(t) = e^t$, $y(t) = e^{-t}$. Find the velocity vector and speed at $t = 0$.
$\vec{v}(t) = \langle e^t, -e^{-t} \rangle$. At $t = 0$: $\vec{v}(0) = \langle 1, -1 \rangle$. Speed $= \sqrt{1^2 + (-1)^2} = \sqrt{2}$.

6 For the particle above, find the magnitude of the acceleration vector at $t = 0$.
$\vec{a}(t) = \langle e^t, e^{-t} \rangle$. At $t = 0$: $\vec{a}(0) = \langle 1, 1 \rangle$. $|\vec{a}(0)| = \sqrt{1^2 + 1^2} = \sqrt{2}$.

Unit tangent vectors

7 For $\vec{r}(t) = \langle t, t^2 \rangle$, find the unit tangent vector $\vec{T}(t)$ at $t = 1$.
$\vec{v}(t) = \langle 1, 2t \rangle$; at $t = 1$: $\vec{v}(1) = \langle 1, 2 \rangle$, $|\vec{v}(1)| = \sqrt{1 + 4} = \sqrt{5}$. $\vec{T}(1) = \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$.

8 Explain why $\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$ always has magnitude 1.
Dividing any nonzero vector by its own magnitude produces a vector with magnitude $\frac{|\vec{v}(t)|}{|\vec{v}(t)|} = 1$ by definition — this is the standard construction of a unit vector in the same direction.

Motion on a circle and centripetal acceleration

9 A particle moves along $\vec{r}(t) = \langle 3\cos t, 3\sin t \rangle$. Find $\vec{v}(t)$, $\vec{a}(t)$, and verify that $\vec{a}(t)$ points toward the origin (is a scalar multiple of $-\vec{r}(t)$).

$\vec{v}(t) = \langle -3\sin t, 3\cos t \rangle$. $\vec{a}(t) = \langle -3\cos t, -3\sin t \rangle = -\vec{r}(t)$. Since $\vec{a}(t)$ is exactly -1 times $\vec{r}(t)$, it points directly toward the origin (centripetal acceleration).

10 For the same particle, verify that $\vec{v}(t) \cdot \vec{r}(t) = 0$ for all t, and interpret this geometrically.

$\vec{v}(t) \cdot \vec{r}(t) = (-3\sin t)(3\cos t) + (3\cos t)(3\sin t) = -9\sin t\cos t + 9\sin t\cos t = 0$. This confirms the velocity vector is always perpendicular to the position vector — consistent with circular motion, where velocity is tangent to the circle.

Displacement vs. distance for vector motion

11 A particle has position $\vec{r}(t) = \langle t^2, t \rangle$ for $t \in [0, 2]$. Find the displacement vector from $t = 0$ to $t = 2$.

$\vec{r}(0) = \langle 0, 0 \rangle$; $\vec{r}(2) = \langle 4, 2 \rangle$. Displacement $= \vec{r}(2) - \vec{r}(0) = \langle 4, 2 \rangle$.

12 Set up (but do not evaluate) the integral for the total distance traveled by the particle $\vec{r}(t) = \langle t^2, t \rangle$ over $t \in [0, 2]$.

$\vec{v}(t) = \langle 2t, 1 \rangle$. Distance $= \int_0^2 |\vec{v}(t)|\, dt = \int_0^2 \sqrt{(2t)^2 + 1^2}\, dt = \int_0^2 \sqrt{4t^2 + 1}\, dt$.

Vector operations review

13 Given $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, find $2\vec{u} - 3\vec{w}$.

$2\vec{u} = \langle 6, -4 \rangle$; $3\vec{w} = \langle -3, 12 \rangle$. $2\vec{u} - 3\vec{w} = \langle 6 - (-3), -4 - 12 \rangle = \langle 9, -16 \rangle$.

14 Find the dot product $\vec{u} \cdot \vec{w}$ for $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, and state whether the vectors are perpendicular.

$\vec{u} \cdot \vec{w} = 3(-1) + (-2)(4) = -3 - 8 = -11$. Since the dot product is not 0, the vectors are NOT perpendicular.

Curvature-related speed problems

15 A particle moves along $\vec{r}(t) = \langle t^3, t^2 \rangle$. Find the times t when the particle's speed is exactly $\sqrt{13}$.

$\vec{v}(t) = \langle 3t^2, 2t \rangle$. Speed$^2 = 9t^4 + 4t^2 = 13$. Let $w = t^2$: $9w^2 + 4w - 13 = 0$. Using the quadratic formula: $w = \frac{-4 \pm \sqrt{16 + 468}}{18} = \frac{-4 \pm 22}{18}$. Taking the positive root: $w = 1$, so $t^2 = 1 \Rightarrow t = \pm 1$.

16 For the particle $\vec{r}(t) = \langle t, \ln t \rangle$ $(t > 0)$, find the velocity vector and speed at $t = 1$.

$\vec{v}(t) = \left\langle 1, \frac{1}{t} \right\rangle$. At $t = 1$: $\vec{v}(1) = \langle 1, 1 \rangle$. Speed $= \sqrt{1^2 + 1^2} = \sqrt{2}$.

Projectile motion as a vector-valued function

17 A projectile is launched with position $\vec{r}(t) = \langle 20t, 30t - 4.9t^2 \rangle$ (meters, seconds). Find the velocity vector $\vec{v}(t)$ and the time when the projectile reaches maximum height (where the vertical velocity component is 0).

$\vec{v}(t) = \langle 20, 30 - 9.8t \rangle$. Maximum height occurs when $30 - 9.8t = 0 \Rightarrow t = \frac{30}{9.8} \approx 3.06$ s.

18 For the same projectile, find the speed at $t = 0$ (launch) and interpret its two components.

$\vec{v}(0) = \langle 20, 30 \rangle$. Speed $= \sqrt{20^2 + 30^2} = \sqrt{400 + 900} = \sqrt{1300} \approx 36.1$ m/s. The horizontal component (20 m/s) stays constant throughout the flight, while the vertical component (30 m/s initially) decreases due to gravity.

Finding when a particle is at rest

19 A particle moves along $\vec{r}(t) = \langle t^3 - 3t,\ t^2 - 4 \rangle$. Find all times t when the particle is momentarily at rest (both velocity components equal 0 simultaneously).

$\vec{v}(t) = \langle 3t^2 - 3, 2t \rangle$. Need $3t^2 - 3 = 0$ AND $2t = 0$ simultaneously. From the second equation, $t = 0$; check the first: $3(0)^2 - 3 = -3 \neq 0$. There is NO time when both components are zero at once, so the particle is never fully at rest.

20 For the same particle, find the times when only the horizontal velocity component is 0.

$3t^2 - 3 = 0 \Rightarrow t^2 = 1 \Rightarrow t = \pm 1$.

Vector-valued functions and arc length

21 State the arc length formula for a vector-valued function $\vec{r}(t) = \langle x(t), y(t) \rangle$ on $[a, b]$, and explain how it relates to the speed $|\vec{v}(t)|$.

$L = \int_a^b |\vec{v}(t)|\, dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$ — arc length is the integral of speed over time, which matches the intuition that distance traveled equals the accumulated speed.

22 Find the arc length of $\vec{r}(t) = \langle 4t, 3t \rangle$ for $t \in [0, 5]$ using the formula above.

$\vec{v}(t) = \langle 4, 3 \rangle$, so $|\vec{v}(t)| = \sqrt{16 + 9} = 5$ (constant speed). $L = \int_0^5 5\, dt = 25$.